THE EXTENDED COCHAIN COMPLEX OF A FAITHFULLY FLAT RING MAP IS EXACT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement refers to the extended cochain complex by label, without displaying it; the display defining it, together with the prose fixing the degrees, is delivered as a context span so that the reader can see the object, and it is not counted as a statement.

Editorial note. The proof contains the author's parenthetical '(proof omitted)' attached to the parallel argument for the cohomology groups in degrees minus one and zero; the argument is a repetition of the displayed reduction and the parenthetical is reproduced verbatim. The step is recorded as an obligation of the card.

Editorial note. The counted spans carry 1 navigation stub; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The extended cochain complex of a ring map and the cosimplicial module it comes from. Same chapter, the display equation-extended-complex defining the complex, delivered as a context span together with the surrounding prose fixing the degrees. Fixes the object whose exactness is asserted.
- Exactness for a ring map admitting a section. Same chapter, Lemma lemma-with-section-exact, invoked at proof lines 798-800. Supplies the statement that the reduction is made to, once R' is chosen to be A.
- Flatness and faithful flatness for ring maps. Algebra, Definition algebra-definition-flat, invoked in the statement at lines 775-776. Fixes the hypothesis and licenses the descent of the vanishing of cohomology along R to R'.

Source reference. Original definition 00HB: sources/stacks-chow/tags.txt, source lines 637–637. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original equation 023K: sources/stacks-descent/pinned-original-source.tex, source lines 729–729. Pointer outside the counted span, invoked once

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Source reference. Original lemma 023L: sources/stacks-descent/pinned-original-source.tex, source lines 741–741. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Suppose that $R \to A$ is faithfully flat, see Algebra, Definition 00HB. Then for any R-module M the extended cochain complex (023K) is exact.*

Proof. Suppose we can show there exists a faithfully flat ring map $R \to R'$ such that the result holds for the ring map $R' \to A' = R' \otimes_R A$. Then the result follows for $R \to A$. Namely, for any R-module M the cosimplicial module $(M \otimes_R R') \otimes_{R'} (A'/R')_\bullet$ is just the cosimplicial module $R' \otimes_R (M \otimes_R (A/R)_\bullet)$. Hence the vanishing of cohomology of the complex associated to $(M \otimes_R R') \otimes_{R'} (A'/R')_\bullet$ implies the vanishing of the cohomology of the complex associated to $M \otimes_R (A/R)_\bullet$ by faithful flatness of $R \to R'$. Similarly for the vanishing of cohomology groups in degrees -1 and 0 of the extended complex (proof omitted).

But we have such a faithful flat extension. Namely $R' = A$ works because the ring map $R' = A \to A' = A \otimes_R A$ has a section $a \otimes a' \mapsto aa'$ and Lemma 023L applies. $\square$